



MINING DEALER BEST PRACTICE

Magnetic Plug for Motor Grader Tandem Case

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

The sprocket/chain drive in a Motor Grader is a large source of fine metal debris in the tandem drive box. A natural property of sprocket/chain drive is that there is rubbing at points of contact, creating small filings. These filings contaminate the lubrication oil and move to other components, which accelerates the normal wear.

This Best Practice outlines the installation of magnetic plugs in the tandem box to aid in the removal of metal particles from the lubrication fluid. The plug should not only keep the oil cleaner, but also reduce component degradation and extra wear.

2.0 Best Practice Description

Currently, only the standard magnetic plug is installed on certain M-series machines. However, to inspect and clean it requires oil in the tandem case to be drained, as it is located in the bottom of the case. This makes regular inspections and cleaning of the magnetic plug only as frequent as the oil-change, preventing it from being effective after time.

Two new magnetic plugs can be installed per tandem box, on the inner side of the cover-plates where the plug would not interfere in the operation of the parts in motion. This location would enable the plugs to be exposed to a considerable amount of oil for cleansing, as it circulates during operation.



Figure 1

The magnetic-plug installed on the cover-plate.



Figure 2

A cover-plate assembly in the tandem box. The plug is clearly not an obstacle.

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3.0 Implementation Steps

1. According to [Appendix 1](#), fabricate the magnetic-plug mounting plate.

Notes:

- a. The mounting plate can be made from 16mm Grade 250 mild steel plate
- b. The magnetic plug (6V-3298) shown in Figures 1-3 are from a D11T Track-Type Tractor. As the plug is the same across most models, and also used in different machines, it is readily available making this Best Practice easier to undertake. See [Appendix 2](#) for an abridged list of machines that use the 6V-3298 magnetic plug.

2. Mark the cover plate with the welding location, and weld the mount to it.

Notes:

- a. On the outer side of the cover plate, it is an option to stamp "MAGNET" into the metal so that technicians are aware of the new service point. This should be done before the welding process to make it easier to stamp the letters.
- b. A thread should be cut into the hole that matches the magnetic plug. The thread size on the plug is 1 $\frac{5}{8}$ "-12.
- c. The mounts should be welded to the back or "inner side" of the cover-plate and be centralised. The weld seams must run parallel to the edges of the cover plate and oriented so that the stem of the magnetic plug faces downward. Figures 1 and 3 shows the orientation relative to the cover plate, and Figure 2 shows orientation relative to the tandem box. Figure 3 shows the cover plate with the breather, where the mounting plate is offset.
- d. Figure 1 shows a cover-plate (8D-1802) from a 16H Motor Grader, which is also shared by the 14H Motor Grader. This Best Practice works for the 16M and 14M Motor Grader models, which have a slightly different cover-plate (266-9728). The same idea can be applied to other Motor Grader models; however, mounting plate dimensions may differ.



Figure 3

Alternative orientation (left) compared to centralized mount (right). Note that the left cover plate also houses the breather.

3. After assuring weld quality, grind the welding seams to give a nice, smooth finish.

- a. Spray-paint the mounting plate for a proper finish.

4. Install the cover plate to the machine.

4.0 Benefits

1. Effective contamination control
2. Cleaner lubrication oil for longer
3. Reduce piston bore scuffing, as seen in brake housing
4. Increase seal and component life

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5. Reduce general wear accelerated by abrasive particles
6. Provides a particle removal component for motor graders not previously installed with a magnetic-plug as standard.
 - Some M series motor graders have a magnetic plug installed from factory, however it is also the drain plug, and requires draining the oil in order to clean and inspect it.

5.0 Resources Required

Fabrication of magnetic plug mount required

Magnetic plug (6V-3298)

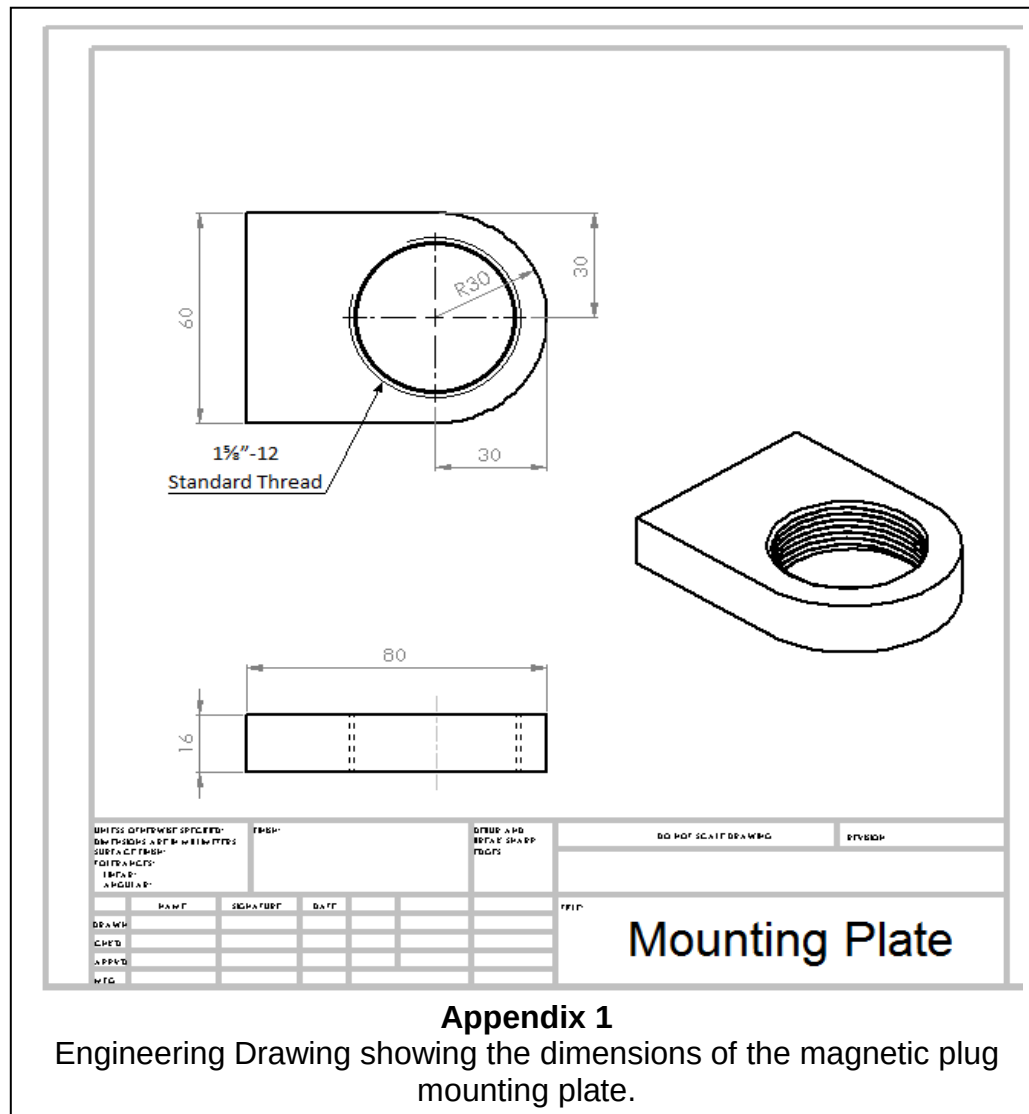
Stamping tools

Appropriate welding tools

Hand grinder

Caterpillar Yellow paint

6.0 Supporting Attachments / References



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Appendix 2
List of machines also using the 6V-3298 Magnetic Plug.

Machine Type	Model	Serial # Prefix
Track-Type Tractors		
D11	N, R, T	N Series: 4HK, 74Z R Series: 7PZ, 8ZR, 9TR, 9XR, AAF T Series: AMA, GEB, JEL, JNS, MDG, TPB
D10	N, R, T	-- Series: 76X, 84W N Series: 2YD, 3SK R Series: 3KR, AKT T Series: RJG
D9	L, N, R, T	L Series: 14Y N Series: 1JD, 6XJ R Series: 7TL, 8BL, ABK, ACL, JJB, WDM T Series: RJS, TWG

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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